Graphic Era HILL UNIVERSITY
Established by an Act of the State Legislature of Uttarakhand (Adhiniyam Sankhya 12 of 2011)
University under section 2(f) of UGC Act, 1956

Mid Term (Odd) Semester Examination October 2024

Roll no.....

Name of the Course and semester: **B.Tech. ME, Semester III**
Name of the Paper: **Mathematics-III**
Paper Code: **TMA 303**
Time: 1.5 hour

Maximum Marks: 50

Note:
(i) Answer all the questions by choosing any one of the sub questions
(ii) Each question carries 10 marks.

1. 10 Marks

a. If $f(z) = u + iv$ is an analytic function and $z = re^{i\theta}$, show that the Cauchy – Riemann equations are $\frac{\partial u}{\partial r} = \frac{1}{r}\frac{\partial v}{\partial \theta}, \frac{\partial v}{\partial r} = -\frac{1}{r}\frac{\partial u}{\partial \theta}$. CO1

Or

b. Show that the function $f(z) = |xy|^{1/2}$ is not analytic at the Origin, although the Cauchy-Reimann equations are satisfied at that point. CO1

2. 10 Marks

a. Find the analytic function $f(z) = u(x, y) + iv(x, y)$ of which the real part $u(x, y) = e^x(x\cos y - y\sin y)$. CO1

Or

b. Show that the function $f(z) = u + iv$, where

$$f(z) = \frac{x^3(1+i) - y^3(1-i)}{x^2+y^2}; \quad z = 0$$

$$f(0) = 0$$

is continuous and that the Cauchy-Riemann equations are satisfied at the origin yet $f'(z)$ does not exist at $z = 0$. CO1

3. 10 Marks

a. Show that $u = \frac{1}{2}\log(x^2 + y^2)$ is harmonic and find its harmonic conjugate. CO1

Or

b. If $f(z) = u + iv$ is analytic function and $u - v = e^x(\cos y - \sin y)$, find $f(z)$ in terms of z. CO1

4. 10 Marks

a. Define bilinear transformation and find the bilinear transformation which maps the points $z_1 = 2, z_2 = i, z_3 = -2$ into the points $w_1 = 1, w_2 = i$ and $w_3 = -1$. CO2

Or

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b. If $f(z)$ is analytic within and a closed contour C, and if a is any point within C, then

$$f(a)=\frac{1}{2\pi i}\int\frac{f(z)dz}{z-a}$$

CO2

10 Marks

5.

a. Using Cauchy's integral formula, evaluate $\oint \frac{\sin \pi z^2+\cos \pi z^2}{(z-1)(z-2)}\,dz$, where C: $|z| = 3$. CO2

Or

b. Discuss the nature of singularity of the function $f(z) = \frac{(z-2)}{z^2}\sin\left(\frac{1}{z-1}\right)$. CO2

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